

Kaiyan Zeng

kaiyanzeng30@gmail.com | 720-382-4046 | www.linkedin.com/in/kaiyanzeng

Green Card holder, no sponsorship needed, willing to relocate

Experience

Necker Fintech

May 2024 – Present

Software Engineer

Denver, CO

- Developed portfolio risk assessment **RESTful APIs** with **Java Spring Boot** serving 200+ holdings, automatically calculating risk scores and generating risk reports based on portfolio and market data, reducing research report data investigation and analysis from **8 hours daily manual work to 2-minute** automated API calls.
- Implemented scalability optimizations including **Redis caching** for frequently accessed risk metrics and asynchronous processing, enabling the system to handle 200+ concurrent requests with average response times **under 500ms**.
- Integrated **TensorFlow-based AI models** via **Python FastAPI** to automatically identify financial anomalies and generate risk commentary sections, enabling analysts to cover **3x more** holdings
- Implemented **Spring Batch overnight jobs** with **MySQL** to pre-calculate daily risk metrics, reducing real-time API computation from **8 seconds to under 1 second** for report generation.
- Configured **OAuth2** authentication with **Spring Security** to protect REST API endpoints, implementing **role-based access control** for research analysts and report reviewers, reducing manual user permission management by **80%**.

Freelancer

Jan 2024 – Apr 2024

Software Engineer

Denver, CO

- Built and deployed email automation backend **RESTful APIs** with **Spring Boot**, secured with **OAuth2 authentication**, to automatically fetch booking data, parsed **JSON** responses with **Jackson**, and triggered personalized confirmation email dispatch, eliminating 50 hours of manual email work monthly.
- Implemented event-driven architecture with **Apache Kafka** to decouple booking events from email processing, using **ThreadPoolTaskExecutor** for parallel execution, reducing customer notification time from **2-4 hours** to under **2 minutes** and enabling system to **handle 500+ emails** monthly without blocking.
- Engineered **fault-tolerant** dispatch logic using **exponential backoff retries** and **SMTP connection pooling**, achieving **99.9% email delivery reliability** despite transient network failures.
- Deployed complete system on **AWS EC2** using **Jenkins CI/CD pipeline** with **Docker containerization**.

Epic Systems

Sep 2022 – Nov 2023

Software Engineer

Verona, WI

- Designed and developed hospital billing automation system serving **50+** healthcare facilities, using **.NET**, **React**, and **Typescript** to automatically select appropriate billing codes based on patient visit data, reducing manual billing code selection time from **30 minutes to 2 minutes** per claim and improving billing accuracy by **15%**.
- Redesigned patient visit **Caché** and **MySQL** schema to support multi-payer insurance and implemented **.NET C# batch processing** service to backfill secondary payer information for **2M+** historical patient visit records, enabling automatic secondary claim generation and eliminating **15,000+** manual claim submissions monthly.

Education

Colorado State University — B.S. in Computer Science, GPA: 3.70/4.00 (2022)

Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, C#, HTML/CSS

Backend & Frameworks: Spring Boot, .NET, Node.js, Spring MVC, RESTful APIs

Frontend: React.js

Databases: MySQL, PostgreSQL, MongoDB, DynamoDB, Redis

Cloud & DevOps: AWS (EC2), Docker, Jenkins, Maven, Gradle, Git, Agile, CI/CD

Testing: JUnit, NUnit, Mockito

Other: Apache Kafka, OAuth2, Spring Security, RBAC